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Heat transfer enhancement in dry cask storage for nuclear spent fuel using additive high density inert gas

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ABSTRACT

Dry cask storage systems (DCSS) is a method of storing high-level radioactive nuclear spent fuel. Due to the decay heat from fission products, effective cooling of the spent fuel is one of the key roles of dry cask storage systems. This study proposes a method to improve the cooling performance of DCSS by adding a small amount of high-density inert gas into helium backfill gas and verifies the method by numerical analysis. First, the candidate group of additive gases that can improve heat transfer in the DCSS environment is selected through figure-of-merit (FOM) analysis based on natural convection heat transfer models. Then, the candidate gases are evaluated using detailed computational fluid dynamics (CFD) modeling without the porous media assumption. From the analysis, it is found that Xe and Kr can reduce the peak cladding temperature (PCT) inside the cask at pressure greater than 1.5 atm. The optimal composition of the additive gas (Xe and Kr) is estimated to be approximately 0.8 in mass fraction. The effect of the additive gases becomes more significant as the pressure increases due to the effect of increased density.

1. Introduction

Spent fuel is nuclear fuel that has been irradiated in a nuclear reactor and is no longer useful for sustaining a nuclear reaction in a nuclear power plant. Spent fuel is often highly radioactive and generates a large amount of decay heat although the fission reaction has ceased (Hubbell, 2011). Although the generation of decay heat will continue to slow down over time, the spent fuel should be cooled and stored securely before being sent for reprocessing or long-term disposal (Romanato, 2011). Spent fuel pool is a good temporary waste storage method designed for low inventories. However, after the Fukushima Daiichi accident, spent fuel pool is considered as a risky option because they are more likely to result in catastrophe owing to ineffective air-cooling for dense-packed fuels in the emergency of water coolant loss. In this circumstance, dry storage systems are widely perceived as a good alternative (Bunn, 2001).

Dry cask storage is a method of storing high-level radioactive spent fuel that has already been cooled in the spent fuel pool for several years. The casks are typically steel cylinders that are either welded or bolted closed. The steel cylinder provides leak-tight confinement of the spent fuel and is surrounded by additional steel, concrete, or other materials to provide radiation shielding. There are various dry storage cask system designs. With some designs, the steel cylinders containing the fuel are placed vertically in a concrete vault; other designs orient the cylinders horizontally. Other cask designs orient the steel cylinder vertically on a concrete pad at a dry cask storage site and use both metal and concrete outer cylinders for radiation shielding (Bunn, 2001).

Effective cooling of the spent fuel is one of the key roles of dry cask storage. Regardless of the cask type, similar heat transfer and cooling processes occur in dry cask storage (Bunn, 2001). In the fuel matrix, heat is produced by radioactive decay and is conducted through the fuel and cladding. Then, the heat is transferred from fuel rods to gaseous coolant by convective heat transfer. Meanwhile, thermal radiation heat transfer occurs between the rows of the fuel rods and between the fuel and surrounding elements. In the internal element and the body walls, heat is transferred to the cooler side by conduction. Eventually, the heat is removed to the environment by natural circulation and thermal radiation at the cask's outer surface.

Previously, various researches have been performed to obtain knowledge of the flow and thermal characteristics inside dry casks. Heng et al. (2002) investigated the natural convection heat transfer in a horizontally oriented dry storage cask in a half-scale 2D model

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Nomenclature

ρ	density [kg/m^3]
μ	dynamic viscosity [kg/ms]
C_p	specific heat capacity [J/kg/K]
k	conductivity [W/m/K]
H	height of the canister [m]
L	length of the canister [m]

Non-dimensional parameters

Ra	Rayleigh number $Ra_L = Gr_L \cdot \Pr = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2} \cdot \frac{c_p\mu}{k}$
Pr	Prandtl number $Pr = \frac{c_p\mu}{k}$
Nu_L	Nusselt number $Nu_L = \frac{hL}{k_f}$

by using PHOENICS-3.2 code. The flow regime inside the canister was close to laminar flow. It was found that as the Rayleigh number increases, the dominant heat transfer mechanism changes from conduction to convection. A recent study was performed using FLUENT code, and a similar conclusion was made by Lee et al. (2013).

Recent studies on concrete storage casks have been actively performed. Lee et al. (2009) investigated the concrete cask model using FLUENT, and half-scale experimental tests in normal and off-normal conditions were compared. The thermal integrity was maintained and the cooling capacity was sufficient even when the flow path was blocked and off-normal condition is applied. Tseng et al. (2011) numerically investigated the thermal performance of a new tube-type dry storage system with 61 BWR spent nuclear fuels using FLUENT. First, the Discrete Ordinates (DO) model in FLUENT was validated though a comparison with the heat flux from an analytical solution on basic components. Second, by comparing the results between the simulation and the experimental tests, CFD was shown to be an effective tool that provides accurate results. Herranz et al. (2015) investigated the effect of some factors (cask design, heat load, and local climate) on the simulation results by comparison with experimental results. A figure-of-merit (FOM) called the margin to safety limit (MSL) was defined for the evaluation of influences of parameters, and heat load was determined to be the most sensitive factor. A more recent study was performed by Wu et al. (2018) on the HI-STORM-100 model using FLUENT. The decay heat of the fuel was calculated from the time when the fuels were loaded in the cask to 50 years after being loaded in intervals of 5 years using ORIGAMI. Using the data, the temperature and flow rate of the outlet of the air channel were calculated and validated.

Yoo et al. (2010) attempted to model detailed representations of fuel rods and claddings without simplification. The porous medium approach and the concept of effective conductivity were excluded to obtain detailed information of the flow and thermal characteristics inside the canister of the TN-24P model. A small model was developed to perform sensitivity studies with a reduced computational cost. FLUENT was used as a tool for numerical analysis, and the results tsts and numerical study (Creer et al., 1987). Brewster et al. (Brewster et al., 2012) conducted thermal and hydraulic tests with a TN-24P dry cask using a unique method. STAR-CCM+code was utilized to simulate the model, which was validated through comparison with a previous study (Creer et al., 1987). To decrease the computational load, a porous media approach was employed. However, the fuel assembly located in the center of canister and expected to show peak cladding temperature (PCT) was explicitly described. The directional calculation of PCT was performed by combining two approaches.

In a dry cask, the backfill gas removes heat from the fuel rod and allows the internal materials to remain chemically stable for long periods of time. For this reason, nitrogen, which is easily obtainable, and helium, which has high thermal conductivity and is inert, are widely used. Previously, a couple of researches have been performed regarding the effect of backfill gas. Li and Liu (2016) performed simulations on a HI-STORM-100 cask with different backfill gases and pressure conditions. Nitrogen showed better thermal performance than helium because of the greater density of nitrogen. In addition, the PCT of the cask was found to decrease as the inner pressure increased. Shin et al. (2016) evaluated the effect of various backfill gases on the thermal performance of the dry cask. Argon had the worst performance because of its low conductivity and specific heat. Helium has high conductivity but showed poor heat transfer performance due to a low buoyancy effect. Nitrogen had the most efficient thermal performance, and air, which is mostly composed of nitrogen, had similar results to nitrogen.

This study proposes a method to improve the heat transfer inside a dry cask using a binary-helium-mixture gas as backfill coolant and evaluates the technical validity using computational fluid dynamics (CFD) analysis. This paper is organized as follows. In chapter 2, a FOM for the cooling performance of natural convection is developed. Based on the FOM, potential mixture gas candidates are screened out, and proper composition ranges are searched. In chapter 3, a numerical model for analyzing the dry storage cask is described. In chapter 4, the candidate gas mixtures and the optimum mixing ratio are discussed based on the results of CFD analysis. In chapter 5, conclusions are made based on further studies.

2. Figure-of-merit (FOM) analysis for helium mixture backfill gas

Previously, several attempts have been made to improve the convective heat transfer performance of gas coolant using binary mixtures. Giacobbe (1998) firstly suggested the improvement of thermal performance in binary mixtures based on a mathematical demonstration using a scaling method in forced convection cases. El-Genk and Tournier (2008) showed that helium-xenon mixtures can enhance performance in closed Brayton cycles and turbines in a VHTR cycle using scaling parameters. Lee et al. (2015) also suggested binary helium-based gaseous mixtures as a new feasible choice of coolant in the fusion reactor blanket. Helium-based gaseous mixtures were proved to reduce circulation power compared to that required with helium. As mentioned above, previous studies have mainly focused on forced convection heat transfer. However, dry cask cooling is controlled by natural convection and radiation heat transfer simultaneously, and therefore, the ideas in previous studies are not directly applicable here. In this study, the effect of radiative heat transfer is ignored in the first screening process because it is not practically achievable to derive an analytical FOM equation by considering both natural convection and radiation heat transfer. In addition, the radiation heat transfer has large geometrical dependency, and it does not allow us to generalize it by only coolant thermal properties. For this reason, this study first screens out the candidate coolants based on the natural convection capability, and then applies detailed CFD analysis for final evaluation.

2.1. Derivation of Figure-of-merit (FOM)

A figure-of-merit (FOM) is a quantity used to characterize the performance of a device, system or method relative to its alternatives (Olivieri and Escandar, 2014). In this study, the derivation of the FOM starts from a natural convection heat transfer correlation in a laminar regime. In a vertical closure with a large aspect ratio, the Nusselt number is calculated as follows (MacGregor and Emery, 1969).

$$Nu_L = 0.42Ra_L^{1/4}Pr^{0.012}\left(\frac{H}{L}\right)^{-0.3}$$

$$\left(10 < \frac{H}{L} < 40, 1 < Pr < 2 \times 10^4, 10^4 < Ra < 10^7\right) \tag{2.1}$$

Eq. (2.1) can be reorganized to be a function of basic fluid properties as follows.

$$h \sim \rho^{0.5}k^{0.738}\mu^{-0.238}C_p^{0.262} \tag{2.2}$$

In this study, we normalized Eq. (2.2) by the reference pure helium gas and defined the result as the FOM. Physically, this FOM indicates the cooling performance of a gas mixture relative to the reference helium coolant for the same geometry and thermal boundary conditions.

$$FOM = \frac{h_{mix}}{h_{He}} = \left(\frac{\rho_{mix}}{\rho_{He}}\right)^{0.5}\left(\frac{k_{mix}}{k_{He}}\right)^{0.738}\left(\frac{\mu_{mix}}{\mu_{He}}\right)^{-0.238}\left(\frac{C_{p,mix}}{C_{p,He}}\right)^{0.262} \tag{2.3}$$

Eq. (2.3) indicates that an additive gas with higher density, heat capacity, thermal conductivity, and lower viscosity is highly preferred. The sensitivity of each parameter on the FOM can be estimated as follows.

$$F_\varphi = \frac{\partial FOM/FOM}{\partial \varphi/\varphi} \tag{2.4}$$

Table 1 shows the calculated sensitivity of each property for the FOM. According to this table, density and conductivity are the most dominant properties that affect heat transfer performance. Generally, as helium has the greatest thermal conductivity among gases in nature except for hydrogen, adding other gases to helium tends to decrease the overall thermal conductivity of the coolant. Therefore, high density is a preferred characteristic for an additive gas to compensate for the thermal conductivity reduction.

2.2. Selection of additive gas candidates

Using the derived FOM, additive gas candidates that could be potentially effective for dry cask backfill gas were evaluated and screened out. The candidates were first selected from among chemically stable gases in the cask environment including nitrogen, carbon dioxide, and inert gases (Ne, Ar, Kr, Xe). The properties of the binary mixture were calculated using the property models as listed in Table 2.

The density and specific heat of the mixture were calculated simply based on mass fraction. The density and specific heat for each species were given as a function of temperature and pressure based on NIST property data. Thermal conductivity and viscosity were calculated by the property model for monoatomic nonpolar molecules. Energy transfer by diffusion was mainly considered rather than energy transfer by collision because the differences in molecular weight between the gases are large in this study. The values of coefficients and exponents were derived from kinetic theory based on ideal gas assumption (Wilke, 1950; Cheung et al., 1962).

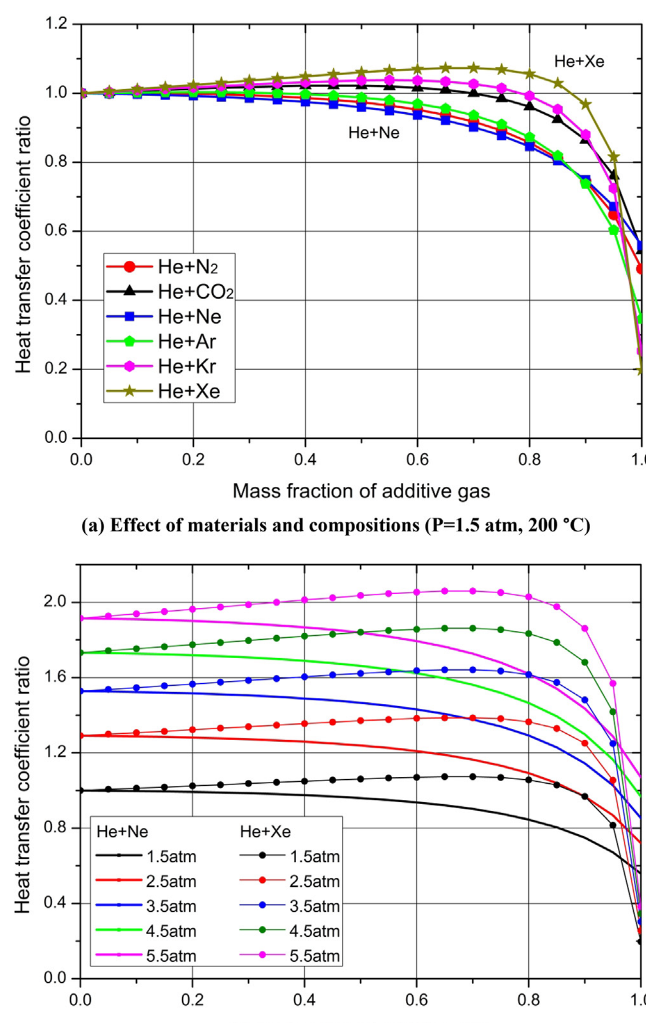
In this study, we calculated the FOMs for gas mixtures of various compositions, and the results are shown in Fig. 1. In this figures, the x-axis is the mass fraction, and the y-axis is the

Table 1
Sensitivity of Parameters on FOM.

Figures of merit	Sensitivity of parameters			
	F_ρ	F_k	F_μ	F_{C_p}
$FOM_h = \rho^{0.5}k^{0.738}\mu^{-0.238}C_p^{0.262}$	0.5	0.738	−0.238	0.262

Table 2
Thermal properties of gaseous mixtures.

Property	Equation
Density (ρ) (Fluent, 2011)	$\rho_{mix} = \frac{1}{\sum_i \frac{Y_i}{\rho_i}} = \frac{\sum_i \rho_i V_i}{\sum_i V_i} = \sum_i \rho_i X_i$
Thermal Conductivity (k) (Cheung et al., 1962; Fluent, 2011)	$k_{mix} = \sum_s \frac{k_i X_i}{\varphi_i}$ $\varphi_s = \sum_j X_j \left[1+\sqrt{\frac{\mu_i}{\mu_j}}\left(\frac{M_j}{M_i}\right)^{\frac{1}{4}}\right]^2 \left[8\left(1+\frac{M_i}{M_j}\right)\right]^{-\frac{1}{2}}$
Viscosity (μ) (Fluent, 2011; Wilke, 1950)	$\mu_{mix} = \sum_s \frac{\mu_i X_i}{\varphi_i}$ $\varphi_s = \sum_j X_j \left[1+\sqrt{\frac{\mu_i}{\mu_j}}\left(\frac{M_j}{M_i}\right)^{\frac{1}{4}}\right]^2 \left[8\left(1+\frac{M_i}{M_j}\right)\right]^{-\frac{1}{2}}$
Heat Capacity (c_p) (Fluent, 2011)	$c_{p,mix} = \sum_i c_p Y_i$



(a) Effect of materials and compositions (P=1.5 atm, 200 °C)

(b) Effect of pressure and compositions

Fig. 1. Comparisons of FOMs.

calculated FOM. The reference pressure for this calculation was selected as 1.5 atm. As shown in Fig. 1(a), the FOM initially increases with mass fraction and then decreases. This means that the effects of conductivity and density are competing depending on the gas composition. However, the trends in the FOM changes were not consistent for different additive gases. For example, the FOM of Xe increases slowly for small mass fractions ($Y_i \leq 0.75$) and then suddenly decrease for large mass fractions ($Y_i > 0.75$). On the other hand, the FOM of Ne decreases gradually as the mass fraction increases. This difference is considered to be due to the molecular weight difference between the two additive gases. The gas with higher molecular weight (Xe) clearly has better potential to increase the heat transfer performance than does the gas with low molecular weight (Ne).

The effect of pressure on the FOM was also investigated. Density is the most influential parameter on FOM, as shown in Table 1. Therefore, an increase in density would have a positive effect on thermal performance. The FOMs of Ne and Xe at pressures greater than 1.5 atm were calculated to determine the relative effect of pressure condition on the FOM in Fig. 1(b). In this figure, the y-axis is the ratio of the heat transfer coefficient of the binary mixture to that of pure helium at 1.5 atm. The enhancement in the heat transfer performance of pure helium relative to that of a binary mixture under the same pressure conditions is shown to increase with increasing pressure.

In the FOM analysis, it was found that that Xe and Kr are the best additive gas candidates in terms of heat transfer. However, other effects such as neutronics and radioactivity should not be ignored for application to dry casks. In the neutronics point of view, Xe-135 is known to be a strong neutron absorber and therefore re-criticality issue can be reduced. In addition, the probability of spontaneous nuclear reactions is low but can weakly occur inside the reactor and neutrons can be generated. Xenon is also a well-known radioactive material, but in reality the isotopes of xenon present in natural world are stable and the isotopes which are radioactive are synthetic, for example Xe-133 and Xe-135. In the case of reactor, the large neutron capture cross section of Xe-135 can be an obstacle but in this situation, there would be no critical problem. Even if Xe-135 absorb neutron, it became Xe-136 which is stable isotopes and decrease of neutrons is not a problem in this situation. In addition, even if other stable isotopes absorb neutron and unstable isotopes are made, those are stabilized through β-decay or electron capture. However, the absorption cross section of xenon isotopes except Xe-135 are very low and the attenuation coefficient of β-ray is very low. Therefore, those effects might not be a critical problem.

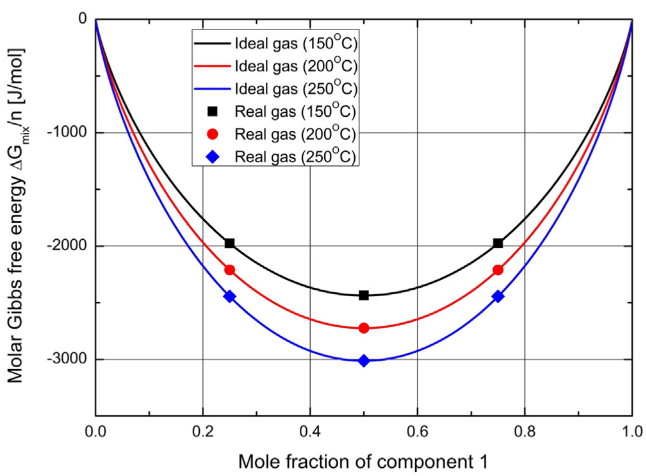


Fig. 2. The change of Gibbs free energy of mixing process.

2.3. Possibility of gas separation and stratification

Although the FOM analysis indicates that Xe and Kr are good additive candidates, there is a concern about separation or stratification between Xe (or Kr) and He gases due to their large density difference. Once they are separated, it can lead to large heat transfer decrease by preventing natural convection. For this reason, it is necessary to confirm if the separation is physically or chemically possible.

In order to discuss whether binary mixture would be separated or not, one can evaluate the spontaneity of the mixing process. In general, the Gibbs free energy is known as an indicator of sponta-

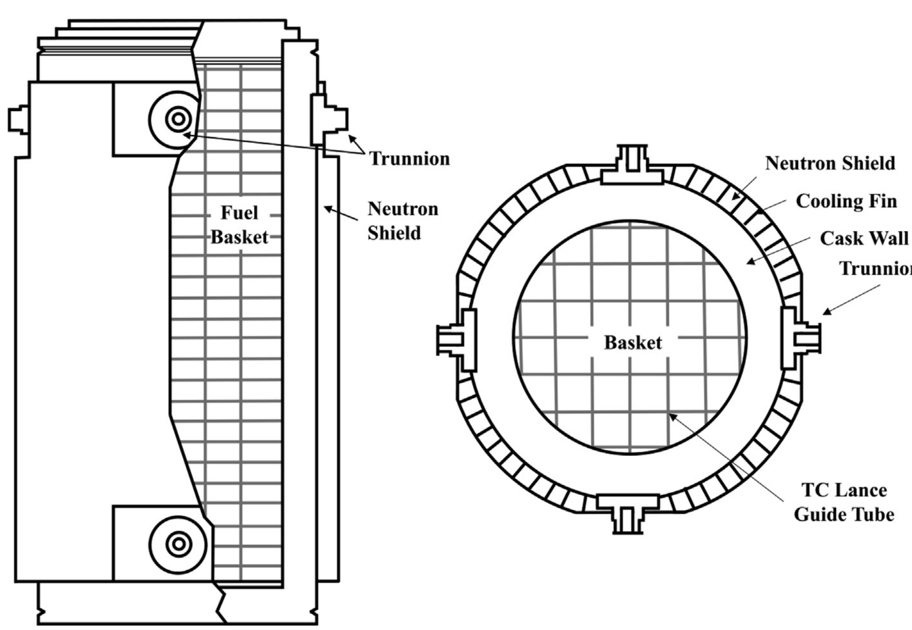


Fig. 3. Schematic design of TN-24P spent fuel dry storage (Creer et al., 1987).

neous process. For ideal gas, the change of Gibbs free energy of the mixing two gases is expressed as follow (Sekerka, 2015).

$$\Delta_{mix}G = nRT(x_1 \ln x_1 + (1-x_1)\ln(1-x_1))(x_1 : \text{component } 1) \quad (2.5)$$

Table 3
The specifications of TN-24P cask (Creer et al., 1987).

Part	Parameters		Values
Body	Height		5063 mm
	Outer diameter		2281 mm
	Loaded weight		100 t
	Material		Forged steel
	Gamma shielding		Steel
	Neutron shielding		Solid
Fuel	Fuel type		PWR W.H. 15 × 15
	Source of assembly		Surry
	Burnup		22-32GWD/MTU
	Cooling time		4.2 years
	Enrichment		2.9–3.2%
	Decay heat	Assembly	832–919 W
		Average	860 W
		Cask total	20.6 kW
Basket	Material		Aluminum
	Criticality control		Borated aluminum/Poison rods

For ideal gases, the Gibbs free energy is always negative and it indicates that the process is spontaneous regardless of the mole fraction. The above equation is derived assuming that intermolecular forces are the same regardless of the molecular species. However, in actual molecules, there is a difference on intermolecular forces between homogeneous and heterogeneous molecules.

$$Z = \frac{V_{real}}{V_{ideal}} = \frac{PV_{real}}{RT} = 1 + \frac{B}{V_{real}} + \frac{C}{V_{real}^2} + \cdots \quad (2.6)$$

The Compressibility factor is the factor that is used to evaluate how far the behavior of a real gas is deviated from the ideal gas behavior and defined as the ratio of the volume of the actual gas to the volume of the ideal gas (Oxtoby et al., 2015). B and C in this term are the second and third virial coefficients, respectively and those are functions of temperature and compositions of gas mixture. Using the virial coefficients, the following equation is derived by taking into consideration the interactions between these molecules (Sattar, 2000).

$$\Delta_{mix}G = RT\sum_i n_i \ln x_i + \sum_i n_i \left[\sum_j \sum_k x_j x_k (2B_{ij} - B_{jk}) - B_{ii}\right] p \quad (2.7)$$

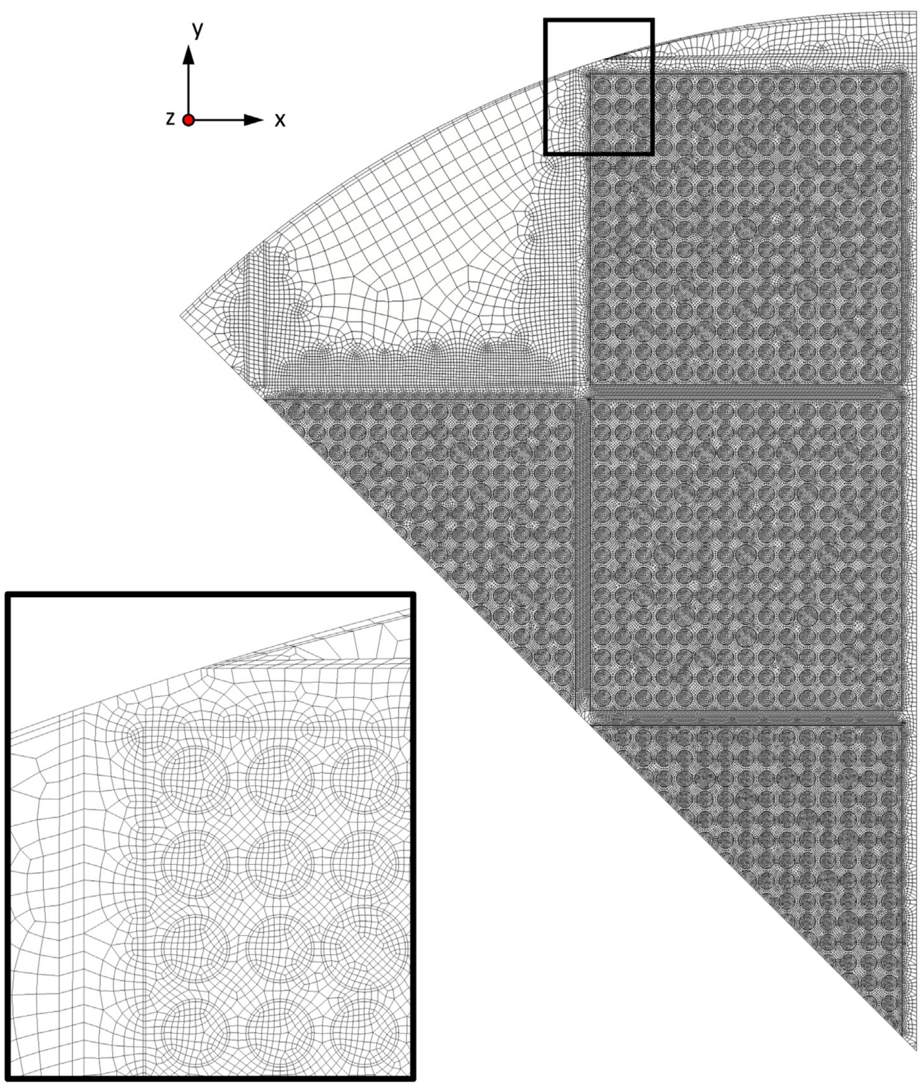


Fig. 4. Grid generation of the TN24P cask model (Yoo et al., 2010).

The parameter B and C accounts for two-body interactions and three-body interactions. The terms beyond second term in Eq. (2.6) were ignored, so this approximation limit the applicability for moderate pressure condition. However, the pressure range we covered is low and the interactions between molecules of the inert gases are not strong. Therefore, the Eq. (2.7) is applicable to our cases. The equation can be simplified as follow for the case of the binary mixture (Sattar, 2000).

$$\Delta_{mix}G = nRT(x_1 \ln x_1 + x_2 \ln x_2) + nx_1x_2(2B_{12} - B_{11} - B_{22})p \qquad (2.8)$$

The first term in right side is same with Eq. (2.5) which is the change of Gibbs free energy of ideal gas and the second term is defined as the excess molar Gibbs energy. This term could be calculated using analytic data with high accuracy of estimation (Kestin, 1984). The changes of Gibbs free energy of ideal gas and real gas at 10 atm are shown in Fig. 2.

The results of the three composition conditions were calculated at temperature range from 150 °C to 250 °C. The excess molar Gibbs energy were far smaller than the Gibbs free energy change of ideal gas because the interaction between molecules of inert gases are small. Therefore, it confirms that binary gases at any temperature and pressure condition will blend each other spontaneously. Also once they are mixed, it is impossible to separate them in a spontaneous process.

3. Numerical model

In the previous section, FOM analysis showed that additive gases (Xe or Kr) with large molecular weight can potentially improve the heat transfer performance inside the cask. However, in reality, the geometry inside the cask is not as simple as assumed

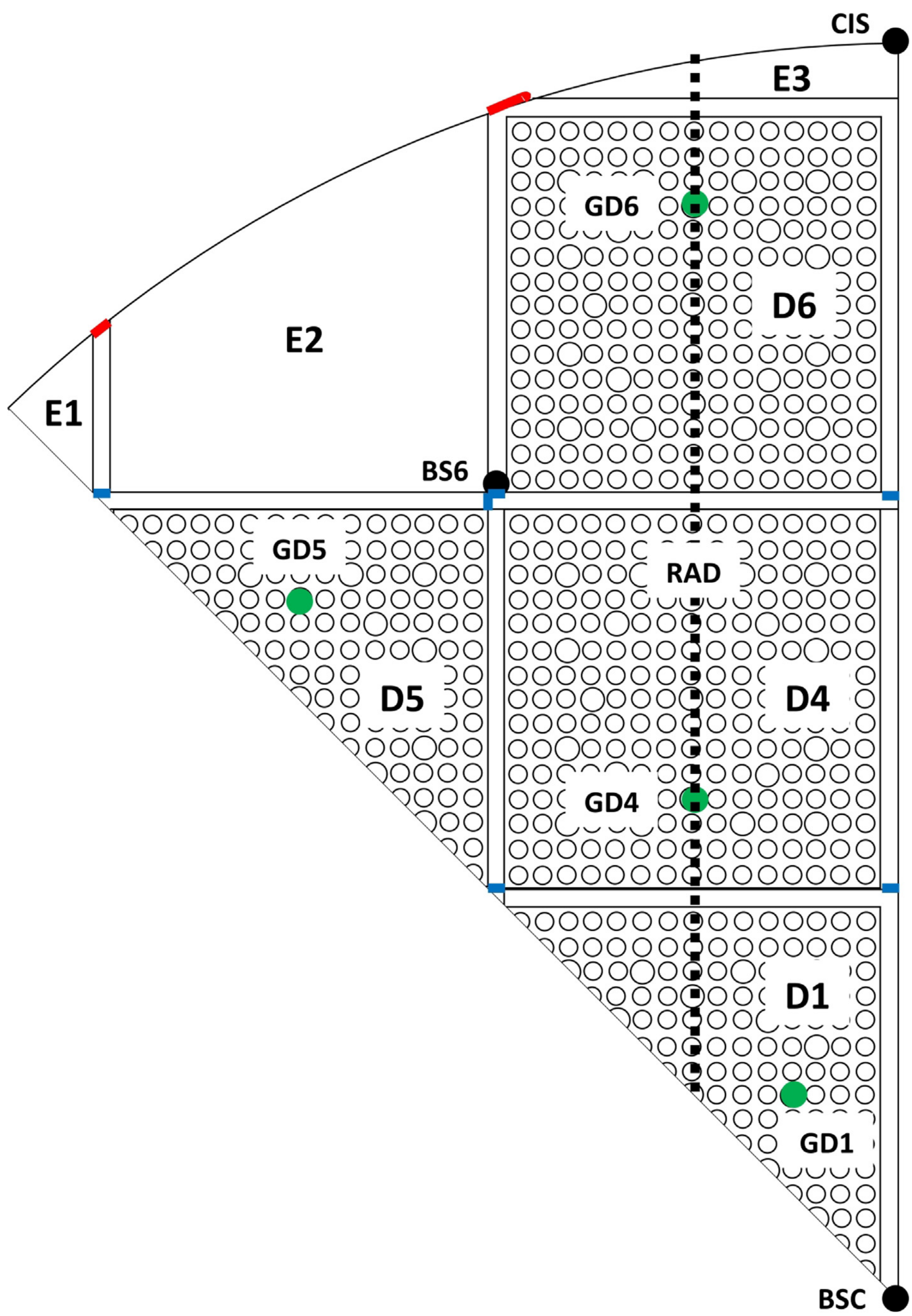
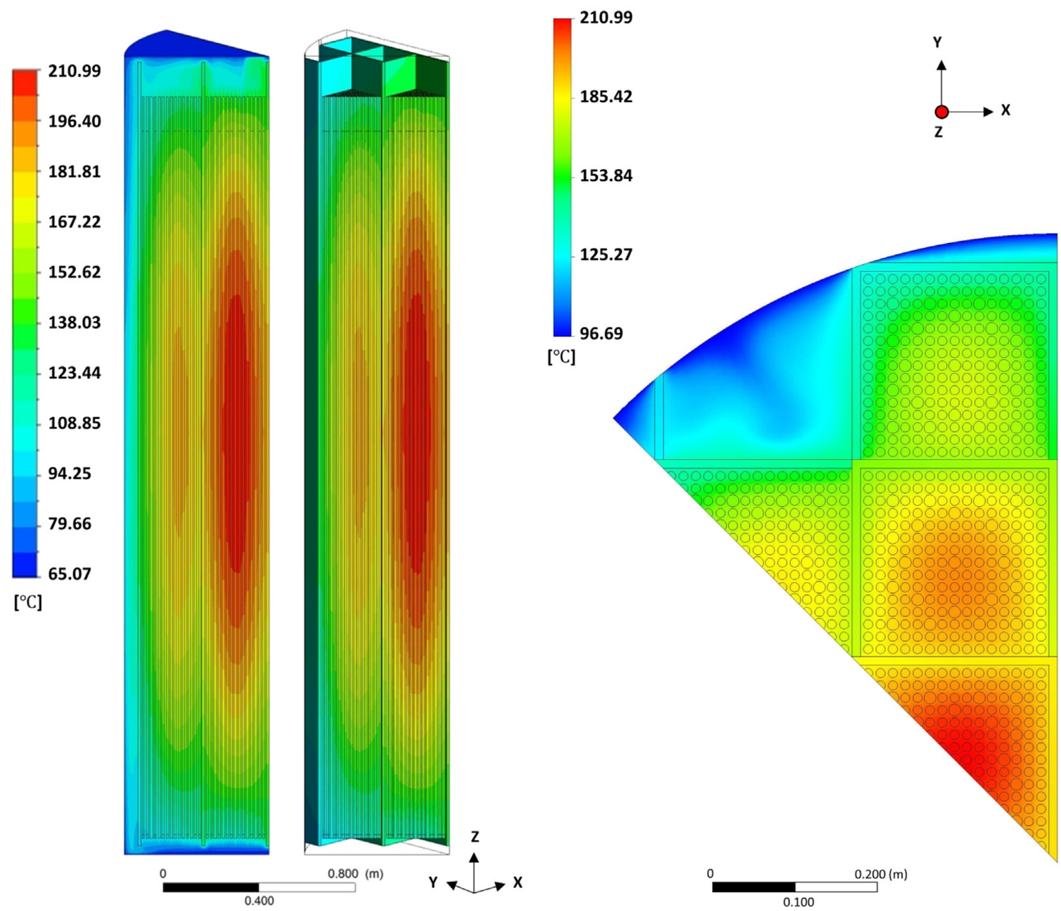
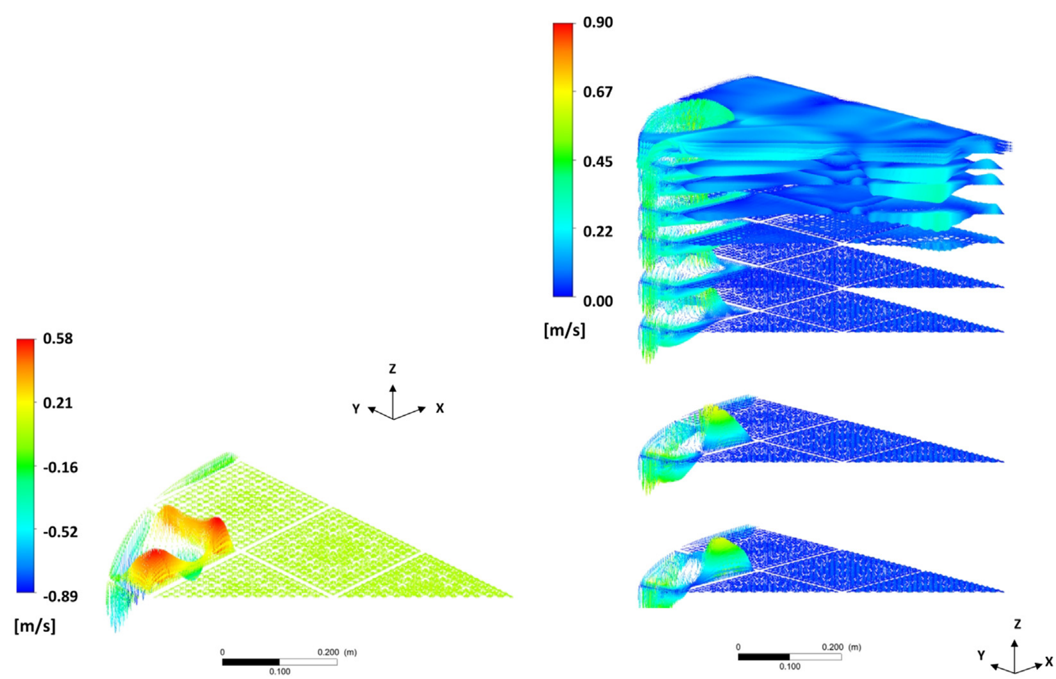


Fig. 5. Locations of spent fuel assemblies and observed temperature (Yoo et al., 2010).



(a) Overall temperature distribution of cask



(b) Flow distribution of the plane at PCT height and upper plenum

Fig. 6. Flow and temperature distribution of TN24P cask.

in the previous section. In addition, the heat transfer mechanism in the cask is governed by natural convection and also controlled by radiation heat transfer, which makes the phenomena more complicated. Therefore, whether the proposed additive gas is effective in a real cask environment and how effective the additive gas is must be evaluated. In this study, we used CFD analysis to achieve this goal.

3.1. Reference Model: TN-24P cask

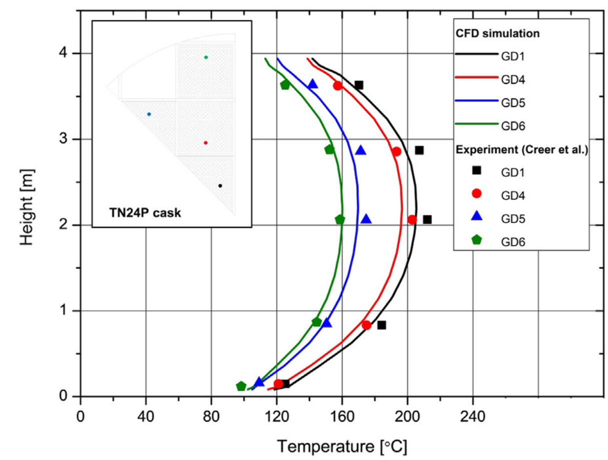
In this study, we selected a TN-24P dry cask, which is a vertical-type system developed by AREVA, as the reference dry cask storage system. The overall geometry and specification of the TN-24P cask are summarized in Fig. 3 and Table 3. The TN-24P dry cask is a one-piece cylindrical forged-steel body surrounded by a resin layer for neutron shielding with a smooth steel outer shell. The body and shell are held together using welded copper angle brackets to provide structural integrity. The fuel basket is loaded with fuel assemblies in the cavity. The basket is an array of interlocking plates, which are composed of a neutron-absorbing material, borated aluminum, to control criticality. Surry spent-fuel assemblies are loaded in the TN-24P cask. The assemblies have standard Westinghouse 15×15 rod designs, which include 20 guide tubes for the control rods and poison rods and 1 instrument tube from among the total 225 rods. The total decay heat in the cask is estimated to be 20.6 kW (Yoo et al., 2010; Creer et al., 1987; McKinnon, 1989).

3.2. Model Set-up

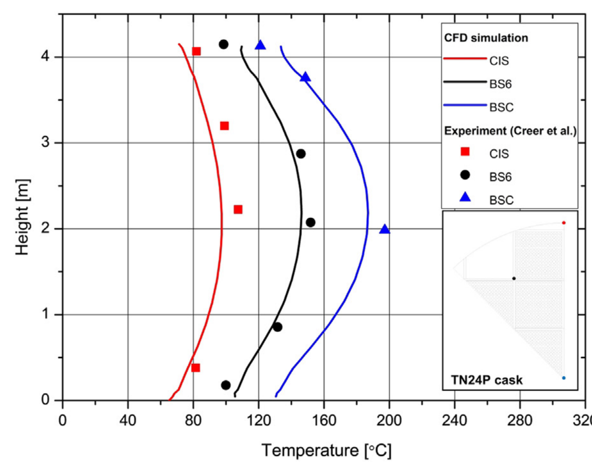
In this study, the cask model was simplified as a 1/8 section to decrease the computational load, and approximately 4.7 million cells were generated. Two entire spent fuel assemblies and two half-size spent fuel assemblies were included in the modeling domain. Previously, most dry cask models used a porous media approach to model the fuel rod assemblies in order to reduce computational cost. However, in this study, we modeled all the fuel rods individually to capture the detailed physics inside the cask. The modeling approach mostly referred to Yoo et al.'s recent CFD models for dry casks (Yoo et al., 2010). The active region and non-active region of the fuel assemblies were modeled separately, and each assembly followed the Westinghouse 15 × 15 rod design. The volumetric heat flux was given for each fuel rod as a heat source. A spacer grid or other instruments in the assemblies were neglected, and a basket composed of boron-aluminum interlocking plates was modeled as separate rectangular plates composed of pure aluminum. Fig. 4 shows the grid model used in this study.

In this study, the flow inside the cask was modeled as a laminar flow. Since the gas density is very low and the coolant channels between the fuel rods are narrow, the flow regime is typically considered as a laminar flow (Heng et al., 2002). The pressure discretization scheme was set to be body-force-weighted to simulate the flow driven by the buoyant force exactly. The discretization schemes of the momentum, energy, and DO radiation model were set as second-order upwind. The discrete ordinates (DO) model was selected to model the radiative heat transfer in the cask. This model is a conservative method that results in a heat balance for coarse discretization. The DO model solves the radiative transfer equation (RTE) for a finite number of discrete solid angles. The theoretical details of the DO model can be found in several references (Fluent, 2011; Yoo et al, 2010).

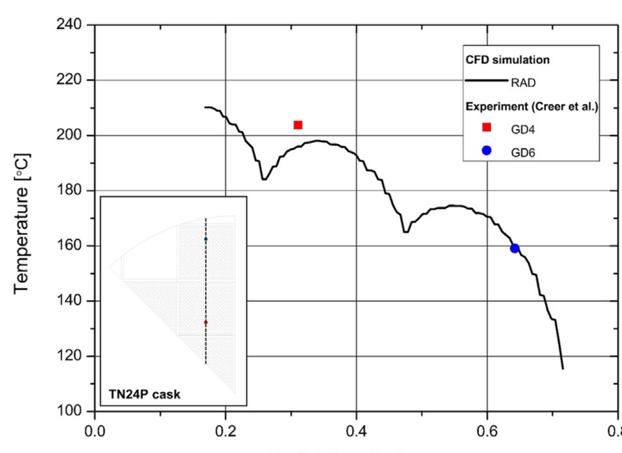
As mentioned above, porous media approaches with effective thermal conductivity have been widely used in previous dry storage cask thermal analysis. However, this study modeled each fuel rod individually without assuming that the competing effect of natural convection and radiation heat transfer would be captured. The fluid properties were calculated using the property models listed in Table 2. In addition, the mass diffusion of the binary mixture was considered in this study. The gas mass diffusivity of the binary mixture was calculated using a formula based on kinetic theory (Fluent, 2011). However, since the mixtures were fully mixed initially, the simulation result was not affected. The surface emittances of the fuel rods and baskets were set equal to 0.8 and the emittance of the canister inner surface was set equal to 0.9,



(a) Temperature distribution of the guide tubes



(b) Temperature distribution of the surfaces



(c) Radial temperature distribution passing the maximum temperature point

Fig. 7. Comparison of temperature distribution between the experiment and numerical results (Yoo et al., 2010).

as in a previous study (Yoo et al., 2010). For the inner surface of the canister, the temperature profile was given as a boundary condition. The basket in the TN-24P cask is composed of stacked interlocking plates (Yoo et al., 2010). Two types of gaps develop when the plates are assembled: the gap between the plates and the gap between the plate and the inner wall. These gaps are filled with backfill gas, and conductive heat transfer is dominant in the gaps because the widths are narrow. In this study, the gap inside the basket was modeled as 0.25 mm, which is the blue line in Fig. 5; the gap between the basket and the inner wall of the cask was modeled as 1 mm, which is the red line in Fig. 5. These model parameters were determined following a previous study (Yoo et al., 2010).

4. Results and discussion

This section summarizes and discusses the results calculated from the numerical model described in the previous section. First, the analysis was conducted for pure helium at 1.5 atm, and the results were compared with reported experimental results (Creer et al., 1987) for model validation. Then, the fluid properties in the model were changed for different additive gases, and the analysis was repeated for various compositions. As discussed in Section 2, Xe and Kr were considered as candidate additive gases.

4.1. Validation of numerical model

For validation of the numerical model, analysis was performed for pure helium coolant. Fig. 6 shows contour plots for the temperature and velocity distribution in the cask. According to Fig. 6(a), the highest temperature occurs near the center of the cask, and the temperature decreases as the location moves to the canister outer surface. Fig. 6(b) shows the velocity profile at the elevation of the peak cladding temperature (PCT) and at the upper plenum of the cask. The helium backfill gas rises slowly inside the baskets where the fuels are located because of large frictional loss. Then, the gas exits the baskets and reaches the upper plenum. The flow from the assemblies are similar to Bénard-Rayleigh flow (Tseng et al., 2011). Then, the gas moves down with relatively high speed in the E1 and E3 regions where the gaps are narrow. Upward and downward flows are observed to coexist in the E2 region.

The calculated results were compared with experimental data in detail and are shown in Fig. 7. Fig. 7(a) shows the axial temperature distribution of the guide tubes in each fuel assembly. Fig. 7(b) shows the axial temperature distribution of the center of the basket, crossing point of baskets 4 and 6, and inner surface of the canister. Fig. 7(c) shows the radial temperature distribution of the line that passes the PCT point and guide tubes (GD4, GD6). All the results show good agreement between the experiment and analysis. In this study, the calculated PCT was 211 °C and showed good agreement with the literature to within 5% (Yoo et al., 2010; Creer et al., 1987).

4.2. Analysis of helium binary mixture backfill gas

In this section, analyses were conducted for the two additive gas candidates (xenon and krypton) using the same numerical model in Section 4.1. Fig. 8 shows the calculated temperature distributions for different mass fractions. In this figure, the location of the PCT moves upward as the mass fraction increases. This is due

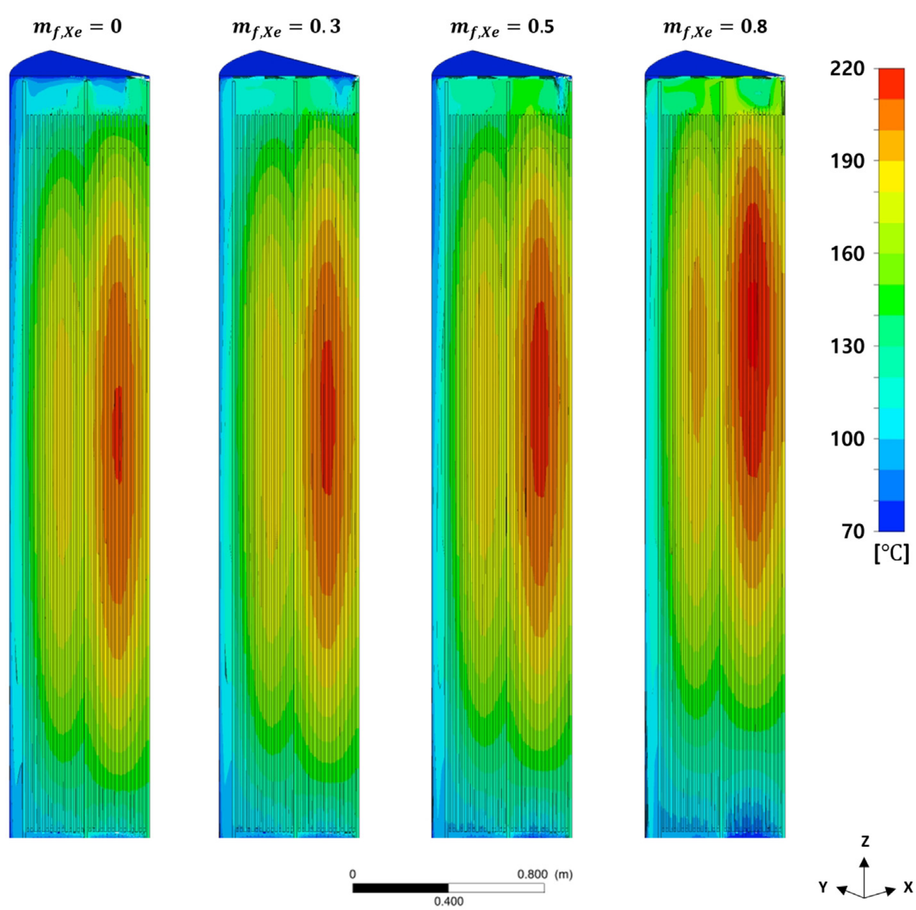


Fig. 8. Effect of mass fraction of Xenon on the height of the PCT.

to the increase in coolant velocity in the heavier coolants. If the mole fraction of the additive gases increases, the density of the mixture gas increases, which results in higher buoyancy forces. This eventually leads to higher natural circulation flow velocity. Fig. 9 shows the changes in PCT with various additive gas mass fractions. As shown in this figure, the PCT in the cask gradually increases with increasing additive gas fraction. This result is not coincident with our expectation that the PCT will decrease with mass fraction due to the enhanced natural convection heat transfer effect. The reason for this discrepancy is thought to be as follows. First, the contribution of conductivity on heat transfer seems to be more significant than that of density unlike the simple FOM analysis. Thus, the increase in density could not compensate for the decrease in conductivity when the mass fraction increases. Second, the effect of radiation heat transfer seems to reduce the effect of natural convection cooling. Therefore, although the coolant properties were significantly changed, they did not positively affect the total heat transfer.

To enhance the density effect, the pressure of the cask was increased, and a similar analysis was conducted. According to the gas property model, the density increases with pressure but the conductivity remains constant. Therefore, higher pressure can enhance natural convection heat transfer without sacrificing conductivity reduction.

Additional calculations with different pressure conditions were carried out to broadly understand the effect of pressure change on the heat transfer performance. The results are shown in Fig. 10. As shown in this figure, when the cask is pressurized at a pressure greater than 1.5 atm, the PCT gradually decreases as heavier gases are added. The minimum PCT is estimated at mass fraction of 0.8 for both Xe and Kr. At mass fractions greater than 0.8, the temperature increases drastically. According to the results, the transition of the dominant heat transfer mechanism is assumed to occur between 1.5 atm and 2.5 atm.

The contribution of radiation heat transfer on cask cooling was also investigated. Fig. 11 shows the ratio of radiation heat transfer to total heat transfer for different pressures and mass fractions. As shown in this figure, the portion of radiation heat transfer is not affected by mass fraction. However, the pressure significantly affected the contribution of radiation heat transfer. When the pressure was 1.5 atm, the dominant heat transfer mechanisms in the cask are conduction. However, if the cask is pressurized at a pressure greater than 1.5 atm, the prevailing heat transfer mechanism changes from conduction to convection. As convection become dominant, the overall temperature decrease and the radiative heat transfer decrease. Therefore, the effect of the increase in convective heat transfer becomes more dominant.

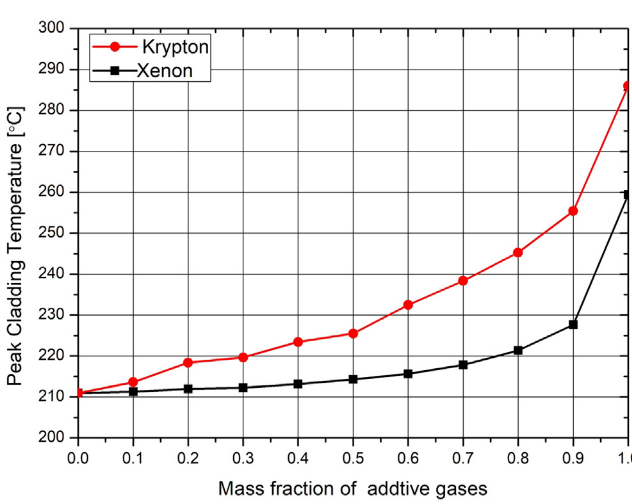


Fig. 9. Effect of mass fraction compositions on PCT (P = 1.5 atm).

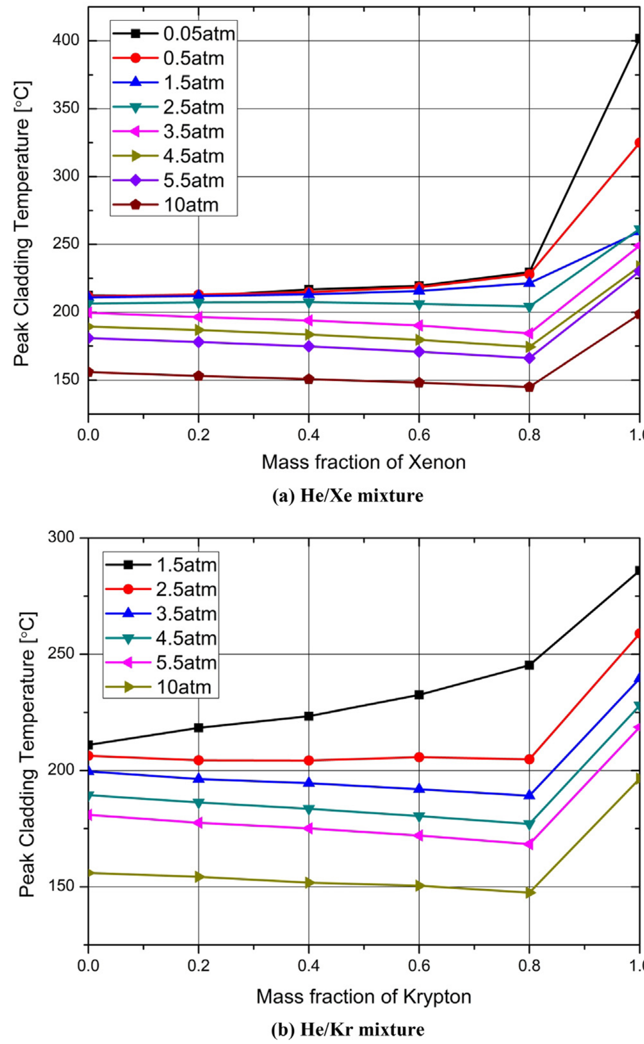


Fig. 10. Effect of pressure and compositions on PCT.

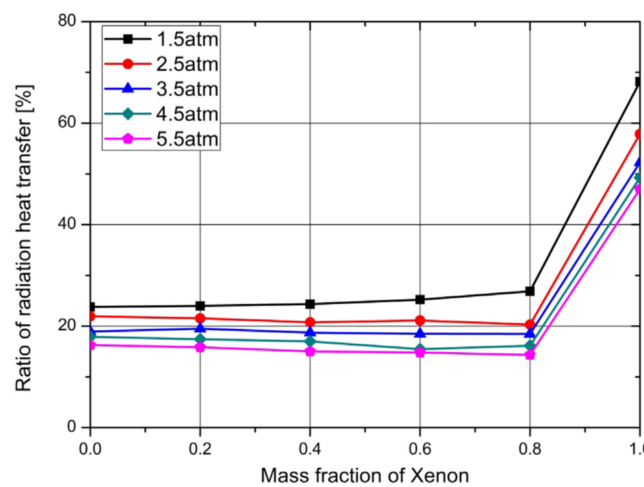


Fig. 11. Effect of pressure and composition on radiative heat transfer proportion of He/Xe mixture.

5. Summary and conclusions

Dry cask storage systems (DCSS) is a method to store high-level radioactive waste such as spent nuclear fuels. Due to the decay heat from fission products, effective cooling of the spent fuel is one of the major roles of dry cask storage. In this study, we propose a method to enhance the cooling performance of a DCSS by adding a small amount of high-density inert gas into helium backfill gas and verify the method through numerical analysis.

In this study, we derived an FOM for DCSS cooling based on natural convection and preliminarily evaluated candidate additive gases. As a result, the FOM was found to increase with mass fraction initially and then decrease. This indicates that the effects of conductivity and density are competing with each other depending on the gas composition. Finally, Xe and Kr were selected as good potential additive gas candidates due to their high molecular weight and chemical stability.

The thermal performance of helium-based binary gaseous mixtures was evaluated in detail based on CFD analysis of a TN-24P dry cask model. In the CFD analysis, enhancement in heat transfer performance was not shown in a cask environment below 1.5 atm in pressure, unlike the results of the FOM analysis, because the effect of conductivity was more significant than that of density. The increase in density did not compensate for the decrease in conductivity as the mass fraction increased. As a result, the PCT increased with the addition of heavier gases. However, a thermal performance enhancement was observed as the pressure increased. According to the gas property model, density increases with pressure, but conductivity remains constant. Therefore, higher pressure could enhance the natural convection heat transfer without a reduction in conductivity. Furthermore, as the overall temperature decreased, the contribution of radiation heat transfer decreased, and the convective heat transfer became more predominant. In this study, the optimal composition of additive gases (Xe and Kr) was estimated to be approximately 0.8 in mass fraction for all pressure ranges investigated.

However, if the cask-inside pressure increases, structural integrity issue cannot be ignored. The conventional dry cask system is generally pressurized to the pressure higher than atmospheric pressure to ensure a certain level of heat transfer capability and to prevent leakage. Generally, the inner pressure differs depending on the type of a storage system and the manufacturer. In the case of TN24 cask, the operating pressure is reported to be around 2.2 atm. In this study, the heat transfer characteristics of binary inert gases with different pressure and composition have been investigated and their optimal conditions have been searched. In the course of analysis, it was found that the heat transfer enhancement by binary mixing can be expected only for the pressure higher than the atmospheric pressure. According to the analysis, Xenon can provide an additional temperature margin of 10–15 °C for 3.5–5.5 atm in pressure. Therefore, in practice, it is important to determine the optimal gas composition by considering all the related factors including thermal, structural, and economical benefits. Further studies are necessary here.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.anucene.2019.04.018.

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